

Rishik Kasula

Charlotte, NC · (704) 421-0449 · contact.rishikkasula@gmail.com · [linkedin](#) · [github](#)

SUMMARY

M.S. in Computer Science (Data Science), UNC Charlotte (May 2026, GPA 3.80), building ML and AI/LLM systems end-to-end. Shipped a calibrated fraud-scoring service (scikit-learn, FastAPI) making expected-cost decisions under bootstrapped, dollar-based evaluation; built an XBRL-graded harness measuring how reliably local LLMs extract financial figures; contributed information-theoretic scoring and clustering to a faculty-led code-analysis research tool. Background in Kafka pipelines, on-prem LLM deployment, and Power BI analytics.

TECHNICAL SKILLS

Machine Learning & AI: Python, scikit-learn, Classification, SVM, Feature Selection, Calibration, Model Evaluation, Bootstrap Confidence Intervals, Mutual Information, Optimization (GA, PSO)

AI & LLMs: LLMs, Retrieval-Grounded Generation, Prompt Engineering, LLM Evaluation Harnesses, Hugging Face, PrivateGPT, On-Prem Deployment, OpenAI API

Data Engineering: Apache Kafka, ZooKeeper, ETL, SQL, Real-Time Ingestion, Data Quality & Validation, Avro, AWS Glue/S3, DuckDB

Engineering & Deployment: FastAPI, Docker, CI (GitHub Actions), Git, pytest, mypy, Power BI, Matplotlib, Linux

EDUCATION

University of North Carolina, Charlotte

Charlotte, NC

Master of Science (M.S.) in Computer Science — Data Science Concentration

GPA: 3.80 | Graduated May 2026

Coursework: Artificial Intelligence, Big Data Analytics, Knowledge Discovery in Databases, Visual Analytics, Database Systems

Vellore Institute of Technology

Amaravati, India

B.Tech in Computer Science (Networking & Security)

May 2024

WORK EXPERIENCE

Graduate Research Assistant — Applied ML & Static Code Analysis

Charlotte, NC

UNC Charlotte, College of Computing & Informatics

Jan 2026 – May 2026

- Designed information-theoretic dynamic feature weighting (Python, 313 LOC) with adaptive discretization for zero-inflated estimates, integrated into the composite scoring layer of *RefExp*, a faculty-led code-analysis research tool.
- Built a multi-level influence-clustering module (~589 LOC) and fixed 3 query and scoring bugs (378 LOC) degrading results across 60+ evaluation rows; credited in the lead RA's integration commit and a file docstring.
- Built pipeline automation (~573 LOC single-command driver) and terminal reporting (~710 LOC); prototyped an LLM-based explanation module on a feature branch (~475 LOC). ~3,500 LOC of git-verified contributions.

Instructional Assistant — Software Engineering for AI-Enabled Systems

Charlotte, NC

UNC Charlotte, College of Computing & Informatics

Sep 2025 – Dec 2025

- Supported graduate instruction on AI system design, data quality, model evaluation, interpretability, and software testing for AI-enabled systems (Prof. Krasniqi, syllabus-documented role); mentored student teams across the ML lifecycle — data collection, cleaning, labeling, feature engineering, training, evaluation.

Intern

India

Inspirisys Solutions Ltd.

Jan 2024 – May 2024

- Deployed PrivateGPT in both local open-source LLM and OpenAI-API configurations on a Windows CUDA/Anaconda stack (Hugging Face model setup) — secure LLM workflows over sensitive banking data.
- Implemented an Apache Kafka + ZooKeeper ingestion pipeline for real-time querying over sensitive banking-IT data (CDC, HDFS, AWS S3, PII handling); built Power BI dashboards on banking-IT KPIs and supported ETL maintenance.

PROJECTS

FraudScore — Calibrated Card-Fraud Scoring Service

github.com/RISHIKKASULA/fraudscore

- Built and shipped (v1.0.0, public repo) a calibrated card-fraud scoring service in Python (scikit-learn, FastAPI, DuckDB, Docker, CI) that decides by expected cost — flagging a transaction when $P(\text{fraud}) \times \text{amount}$ exceeds the review cost.
- Under leak-free chronological evaluation, an amount-aware rule saved \$84.87 per 10k transactions (95% CI \$20.08–\$187.47), roughly 13% cheaper than the best single-threshold baseline; found a plain logistic model beat gradient boosting, with every headline number bootstrapped.

filings — Evaluation Harness for LLM Extraction from SEC Filings

public repo, v0.1.0

- Built and released a harness that grades local-LLM extractions automatically against each company's own XBRL filings — a 600-call deterministic grid (10 filers $\times$ 5 KPIs $\times$ 3 local models $\times$ 2 $\times$ 2 ablation), reported with cluster-bootstrap confidence intervals by company.
- Measured a model-size cliff on headline-figure extraction with the financial statements in context: an 8B local model reached 87.8% (95% CI 81.5–93.8) vs. 50.5% and 44.4% for the 7B and 3B models (that pair's gap inconclusive); hand-labeled all 230 failures into a six-label taxonomy, tracing a single-KPI collapse to mis-tagged scale (32 of 37).

tickflow — Streaming Data-Quality Gate for Kafka-Compatible Streams (v0.9, in progress)

github.com/RISHIKKASULA/tickflow

- Building an open-source (MIT) contract-enforcement layer for Kafka-compatible streams: per-topic contracts (Avro plus six semantic rules in YAML) enforced inline, violations quarantined with self-describing envelopes. Sole author, 4,411 lines of Python source (7,047 with tests); 194 tests, 99.4% core-module coverage, mypy strict, CI on every push.
- Graded the engine against a seeded fault-injection fixture with a ground-truth manifest of 1,677 labeled faults: replaying it with gates on vs. off cut violated bars from 1,076 to 0 of 15,061, gated output bit-identical to ground truth (fixture-scale).

Feature Selection using Hybrid GA-PSO — Soft Computing, VIT-AP

- Implemented a hybrid Genetic Algorithm–Particle Swarm Optimization feature-selection pipeline (Python, scikit-learn) on remote-sensing data (Indian Pines, 224 bands; Massachusetts Roads), scored by linear-SVM accuracy and Cohen's kappa; converged faster to lower-error subsets than GA or PSO alone.

AI Express Pay System — VIT-AP Capstone

- Built an automated retail-billing system — Raspberry Pi with an Edge Impulse offline vision model, full-stack Node.js/Python app, QR-based UPI payment.

CERTIFICATIONS & LEADERSHIP

Certifications: DeepLearning.AI Mathematics for ML & Data Science (Feb 2025); AWS Academy Data Engineering (Apr 2025, 40 hrs); IBM Python for Data Science, AI & Development; IBM/Cognitive Class Docker Essentials (Apr 2025)

Leadership: Organizing committee, Mana Charitable Trust (2 yrs); Team Lead, VTAPP — 15+ students at VIT-AP's tech fest.